

Week 3 Lab 1 Report

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Task 1:

A global variable `delay_time` was added and initialized at 4000, and divided by 2 at the end of `loop()`.

Task 2:

The pin was changed from D4 to D3. `millis()` and timer variables was used to check the time interval of the two LEDs independently, where the state of each LED was toggled after they passed their time intervals. The following is an example for one of the LEDs:

```
if (millis() - LED_timer > LED_interval) {  
    LED_state = !LED_state;  
    digitalWrite(LED, LED_state);  
    LED_timer = millis();  
}
```

Task 3:

A check in `loop()` is added to toggle the direction of brightness change at the maximum and minimum values of brightness (0 and 255), and the loop uses `analogWrite` for the brightness level onto the pin.

```
if (Brightness >= MAX_BRIGHT)  
    increment = -1;  
else if (Brightness <= MIN_BRIGHT)  
    increment = 1;  
Brightness += increment;
```

Task 4:

A debounce time interval was added and checked, making sure the difference between the current time and the previous time the switch was pressed was above the threshold. If this is true, it toggles the switch. The delay was set `debounceDelay = 200`.

```
if (digitalRead(SWITCH) == LOW  
    && millis() - lastDebounceTime > debounceDelay) {  
    state = !state;  
    lastDebounceTime = millis();  
}
```

Task 5:

```
IRAM_ATTR void toggle() {  
    pressed = HIGH;  
}
```

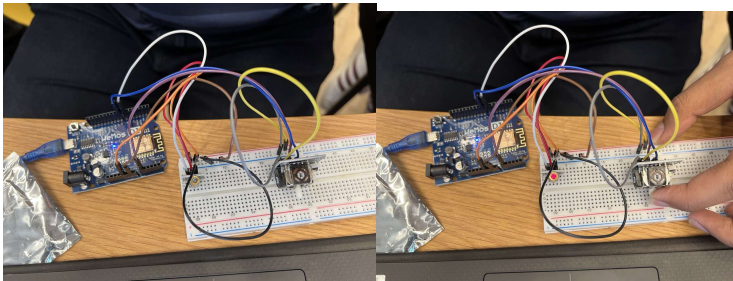
The ISR only changes the variable `pressed`. Inside `loop()`, the `pressed` byte is checked and “cleared”. The debouncing logic is placed afterward.

```
if (pressed == HIGH) {  
    pressed = LOW;  
    if (millis() - lastDebounceTime > debounceDelay) {  
        state = !state;  
        lastDebounceTime = millis();  
    }  
}
```

Task 6:

`temp_state` and `humid_state` bytes were added to check whether the temperature and humidity values were above a threshold. If either states were – for example, the temperature `temp_state = event.temperature > TEMP_THRESHOLD` - then the LED is set to turn on, otherwise, if neither states were, the LED is set to turn off.

Task 7:



The joystick was used to control the laser, using a simple logic check as shown below. The board only had one analog pin A0, so that was connected to the x axis value from the joystick.

```
int x = analogRead(X_PIN);  
Serial.print(x);  
Serial.print("\n\n");  
if (x < 128)  
    digitalWrite(LASER, HIGH);  
else  
    digitalWrite(LASER, LOW);
```